MILITARY SPECIFICATION SHEET

BATTERY, STORAGE, AIRCRAFT, NICKEL-CADMIUM, CELL

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the cell described herein shall consist of this document and the latest issue of specification MIL-B-81757.

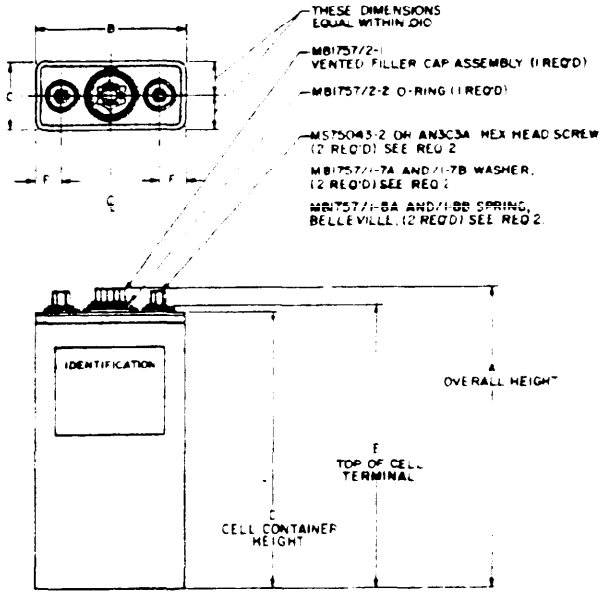


FIGURE 1. Cell assembly.

TABLE I.

Part Number	Used With Battery P/N	Cell Dimensions (Fig. 1) A Max	B +.015 -.025	C +.015 -.025	D Ref	E ±.020	F ±.010	Cell Rated Capacity (AH) Ref	Cell Max Wt Lbs
M81757/1-2	M81757/7-2 M81757/7-3	6.950	2.414	1.065	6.280	6.525	0.395	10	1.30
M81757/1-4	M81757/8-2 M81757/8-3 M81757/11-1 M81757/11-2	8.278	3.175	1.070	7.606	7.869	0.650	20	2.30
M81757/1-6	M81757/9-2 M81757/9-3 M81757/13-1	9.400	3.130	1.375	8.678	8.932	0.540	30	3.40

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TABLE II.

Used with Assembly No.	Terminal Screw	
	Size	Part No.
M81757/1-2 M81757/7-3 M81757/10-1	8-32	MS75043-2 M81757/1-9 M81757/1-9
M81757/1-4 M81757/1-6	10-32	AN3C3A
M81757/8-3 M81757/9-3 M81757/11-1 M81757/11-2 M81757/12-1 M81757/12-2 M81757/13-1	10-32	M81757/1-10

MATERIAL: STAINLESS STEEL. DIMENSIONS ARE IN INCHES.

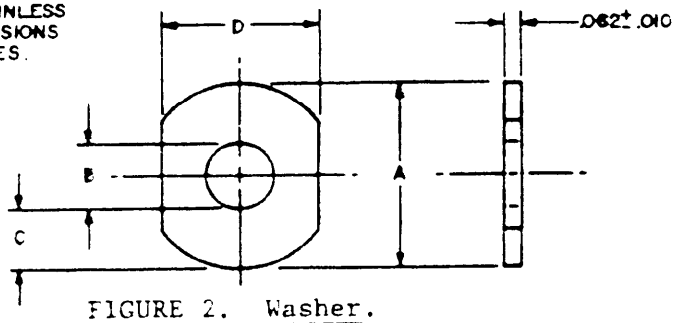


FIGURE 2. Washer.

TABLE III.

Part No.	Used with Assembly No.	A ±.030	B ±.010	C ±.020	D ±.030
M81757/1-7A	M81757/1-2 M81757/7-2 M81757/7-3 M81757/10-1	.500	.178	.161	.400
M81757/1-7B	M81757/1-4 M81757/1-6 M81757/8-2 M81757/8-3 M81757/9-2 M81757/9-3 M81757/11-2 M81757/11-1 M81757/12-1 M81757/12-2 M81757/13-1	.750	.203	.273	.600

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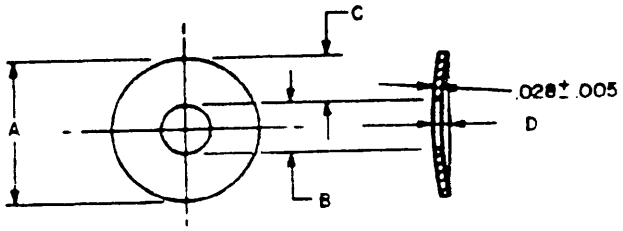


FIGURE 3. Spring, belleville.

TABLE IV.

Part No.	Used With Part No.	A ±.030	B ±.010	C ±.020	D ±.005
M81757/1-8A	M81757/1-2 M81757/7-2 M81757/7-3 M81757/10-1	.500	.178	.161	.011
M81757/1-8B	M81757/1-4 M81757/1-6 M81757/8-2 M81757/8-3 M81757/9-2 M81757/9-3 M81757/11-1 M81757/11-2 M81757/12-1 M81757/12-2 M81757/13-1	.561	.203	.179	.025

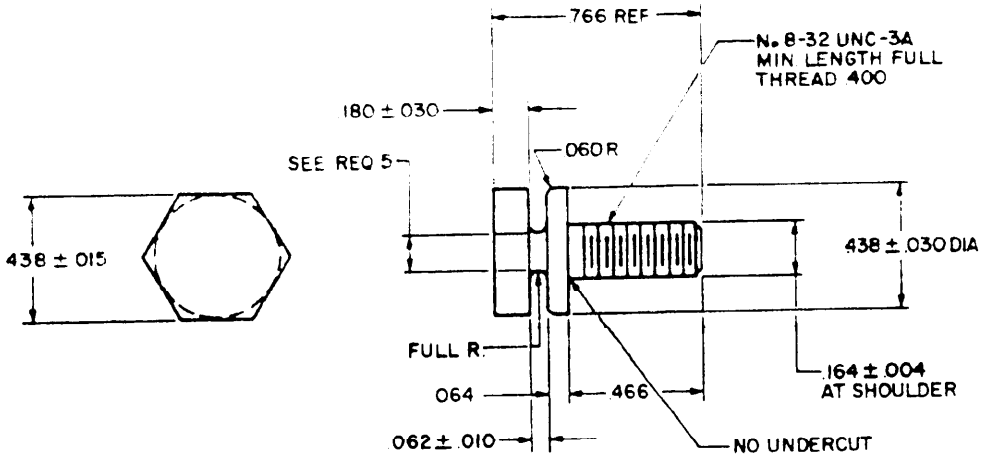
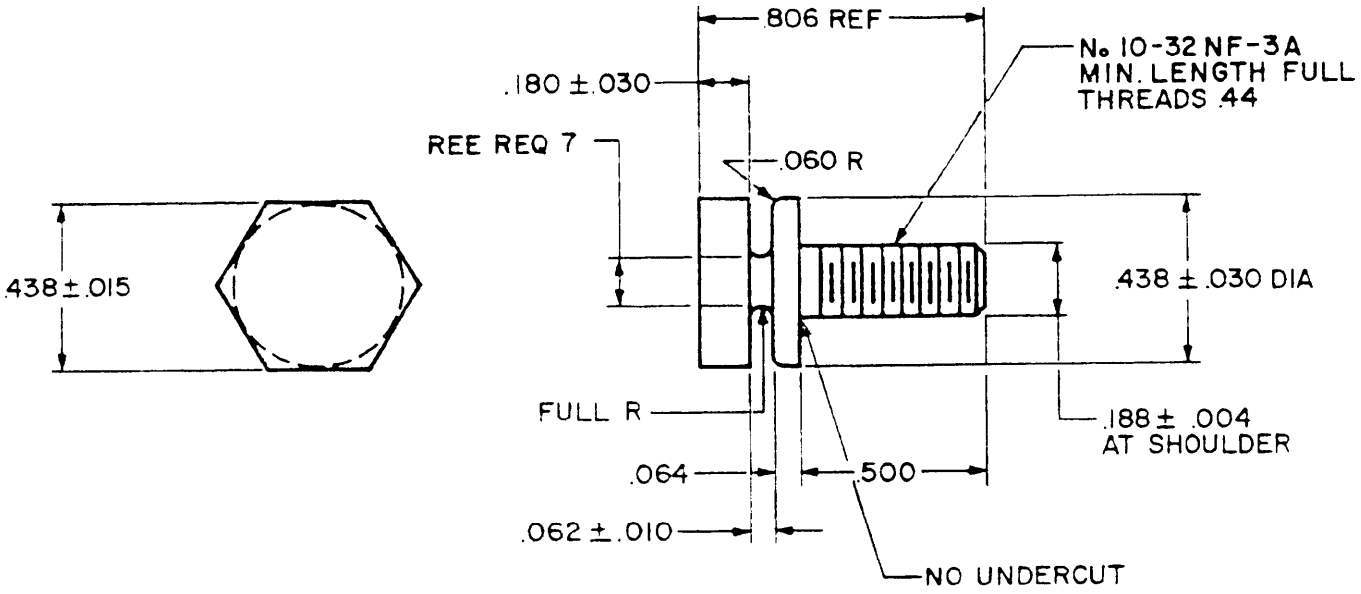


FIGURE 4. Terminal screw.
P/N M81757/1-9

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(C) FIGURE 5. Terminal screw.
P/N M81757/1-10

REQUIREMENTS:

1. Terminal seal material shall be ethylene propylene rubber.

2. Terminal screws, washers and belleville springs are not required for cells that are procured separately from complete batteries.

3. The electrolyte in each cell shall be 0.25 inch above the baffle 2-4 hours after initial charge termination.

(C) 4. Compliance with cell dimensional requirements shall be determined with the cell in a fully discharged state.

(C) 5. The M81757/1-9 undercut diameter shall be 0.112 ± 0.002 between the hex head and washer face.

(C) 6. The M81757/1-9 head break torque shall be 38 to 48 in. lbs and the thread break torque shall be 52 to 68 in. lbs.

(C) 7. The M81757/1-10 undercut diameter shall be 0.128 ± 0.002 between the hex head and the washer face.

(C) 8. The M81757/1-10 head break torque shall be 55 to 72 in. lbs and the thread break torque shall be 90 to 120 in. lbs.

(C) 9. The terminal screw material shall be stainless steel type 303 or equal with a tensile strength of 110,000 to 150,000 PSI.

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NOTES:

1. All dimensions are in inches unless otherwise specified.

(C) 2. Supersession data: This specification sheet supersedes MS90321, MS27307 and MIL-B-81757 part numbers as follows, when a product has been qualified for each MIL-B-81757 part number respectively.

Cell Part Number	Replaces Part Numbers
M81757/1-2	MS90321-66, MS90321-72, MS27307-1, MS27307-4 & M81757/1-1
M81757/1-4	MS90321-68, MS90321-74, MS90321-77, MS90321-78, MS27307-2, MS27307-5 & M81757/1-3
M81757/1-6	MS90321-69, MS90321-75, MS27307-3, MS27307-6 & M81757/1-5

Cell Part Number	Superseded by
MS27307-1	M81757/1-2
MS27307-2	M81757/1-4
MS27307-3	M81757/1-6
MS27307-4	M81757/1-2
MS27307-5	M81757/1-4
MS27307-6	M81757/1-6
MS90321-66	M81757/1-2
MS90321-68	M81757/1-4
MS90321-69	M81757/1-6
MS90321-72	M81757/1-2
MS90321-74	M81757/1-4
MS90321-75	M81757/1-6
MS90321-77	M81757/1-4

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Cell Part Number	Superseded by
MS90321-78	M81757/1-4
M81757/1-1	M81757/1-2
M81757/1-3	M81757/1-4
M81757/1-5	M81757/1-6

International Standardization. Certain provisions of this specification sheet are the subject of International Standardization Agreement (ASCC AIR STD 12/15 and STANAG 3514). When revision or cancellation of this specification sheet is proposed which will affect or violate the international agreement concerned, the preparing activity will take appropriate action through international standardization channels, including department standardization offices, if required.

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Navy - AS
Air Force - 11
Army - ER

Review Activities:
Air Force - 99
Army - AV

User Activity:
Coast Guard - CG

Preparing Activity:
Navy - AS
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